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Electric wireless charging device

Abstract

Provided is an electric wireless charging device, including a charging dock, a rotating charging base, and a first driving component. The first driving component includes a driving body and a driving shaft; the driving shaft is connected with the charging dock, and the driving body is connected with the rotating charging base; and the driving body is rotatably connected with the driving shaft, and the rotation of the driving body relative to the driving shaft is suitable for driving the rotating charging base to rotate. In this way, the electric wireless charging device may charge different electronic devices respectively through the charging dock and the rotating charging base. Furthermore, the driving body may drive the rotating charging base to rotate, whereby the rotating charging base may rotate automatically to an appropriate position, which is convenient for a user to place the electronic device in an appropriate position for charging.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The application claims priority to Chinese patent application No. 2025200300476, filed on Jan. 7, 2025, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of wireless charging devices, and in particular, to an electric wireless charging device.

BACKGROUND

(3) A wireless charging device achieves wireless transmission of electric energy by using electromagnetic induction, magnetic resonance, or electric field coupling, and the like. The wireless charging devices are connected without wires or cables, and can transmit the electric energy to an electronic device from the wireless charging device by means of electromagnetic wave propagation so as to realize wireless charging, and improve the charging convenience of the electronic devices.

(4) However, the wireless charging device in related technologies can charge only one electronic device, has a single function, and cannot meet use requirements of users.

SUMMARY

(5) Embodiments of the present disclosure provide an electric wireless charging device, to solve the above technical problems.

(6) The present disclosure is implemented through the following technical solutions:

(7) Embodiments of the present disclosure provide an electric wireless charging device. The electric wireless charging device includes a charging dock, a rotating charging base, and a first driving component, where the first driving component includes a driving body and a driving shaft; the driving shaft is connected with the charging dock, and the driving body is connected with the rotating charging base; and the driving body is rotatably connected with the driving shaft, and the rotation of the driving body relative to the driving shaft is suitable for driving the rotating charging base to rotate.

(8) In some embodiments, the charging dock includes a fixed mount, the fixed mount is provided with a fixed part, the driving shaft is installed on the fixed part, the rotating charging base includes a rotating support, the rotating support is provided with a connecting part, the driving body is installed on the connecting part, and the rotation of the driving body relative to the driving shaft is suitable for driving the rotating support to rotate.

(9) In some embodiments, the fixed mount is further provided with a guiding part, the rotating support is provided with a guiding mating part, and the guiding mating part is slidably connected with the guiding part.

(10) In some embodiments, the guiding mating part is provided with a wire through hole, and the wire through hole penetrates through the guiding part.

(11) In some embodiments, the charging dock further includes a limit stop and guiding members, the limit stop is connected with the fixed mount, the limit stop is provided with a limit part, the guiding members penetrate through the limit part and are connected with the rotating support, an end, facing away from the rotating support, of each guiding member is limited in the limit part, the guiding members are suitable for limiting the movement of the rotating support along an axial direction of the guiding members, and the guiding members and the rotating support rotate synchronously around a rotating axis of the driving body.

(12) In some embodiments, there are a plurality of guiding members, the plurality of guiding members are disposed around the periphery of the driving body, and the plurality of guiding members are distributed symmetrically relative to the rotating axis of the driving body.

(13) In some embodiments, the electric wireless charging device includes a first charging part and a second charging part, the first charging part and the second charging part are both installed on the charging dock, and the first charging part and the second charging part are located on two opposite ends of the charging dock respectively.

(14) In some embodiments, the electric wireless charging device includes a second driving component, the second driving component includes a driving main body and a driving gear, the first charging part is provided with a first toothed part, the second charging part is provided with a second toothed part, the first toothed part is engaged with one end of the driving gear, the second toothed part is engaged with the other end of the driving gear, the driving main body is suitable for driving the driving gear to rotate, the rotation of the driving gear is suitable for driving the first toothed part to move towards a first direction, the rotation of the driving gear is further suitable for driving the second toothed part to move towards a second direction, where the first direction and the second direction are different.

(15) In some embodiments, the charging dock is provided with a first limit guiding part and a second limit guiding part, the first charging part is provided with a first guiding sliding part, the first guiding sliding part is slidably connected with the first limit guiding part, the second charging part is provided with a second guiding sliding part, and the second guiding sliding part is slidably connected with the second limit guiding part.

(16) In some embodiments, the electric wireless charging device includes the first charging part and

a watch magnetic member, the first charging part is provided with an accommodating cavity, and the watch magnetic member is installed in the accommodating cavity in a reversible manner.

(17) According to the electric wireless charging device provided by the embodiments of the present disclosure, the first driving component of the electric wireless charging device includes the driving body and the driving shaft, the driving shaft is connected with the charging dock, the driving body is connected with the rotating charging base, the driving body is rotatably connected with the driving shaft, and the rotation of the driving body relative to the driving shaft is suitable for driving the rotating charging base to rotate. In this way, the electric wireless charging device may charge different electronic devices respectively through the charging dock and the rotating charging base, which is helpful to enhance functionalities of the electric wireless charging device. Furthermore, the driving body may drive the rotating charging base to rotate, whereby the rotating charging base may rotate automatically to an appropriate position, which is convenient for a user to place the electronic device in an appropriate position for charging, and eliminates the need for the user to manually toggle the rotating charging base, thereby enhancing user operation.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In order to more clearly illustrate the technical solutions in embodiments of the present disclosure, drawings used in the description of the embodiments are simply described below. Apparently, the drawings in the following description are some embodiments of the present disclosure. For those skilled in the art, other drawings can be obtained without creative effort according to these drawings.
- (2) FIG. 1 is a schematic structural diagram of an electric wireless charging device according to an embodiment of the present disclosure.
- (3) FIG. 2 is a schematic structural diagram of the electric wireless charging device of FIG. 1 in another state.
- (4) FIG. 3 is a schematic structural diagram of a first driving component, a fixed mount, and a rotating support of the electric wireless charging device of FIG. 1.
- (5) FIG. 4 is a schematic structural diagram of a first driving component, a fixed mount, a rotating support, a limit stop, and guiding members of the electric wireless charging device of FIG. 1.
- (6) FIG. 5 is a schematic structural diagram of a rotating support, a limit stop, and guiding members of the electric wireless charging device of FIG. 1.
- (7) FIG. 6 is a schematic structural diagram of a second driving component, a first charging part, and a second charging part of the electric wireless charging device of FIG. 1.
- (8) FIG. 7 is a schematic structural diagram of a partial structure of the electric wireless charging device of FIG. 1 in a disassembled state.
- (9) FIG. 8 is a schematic structural diagram of the first charging part of FIG. 6 and a watch magnetic member.
- (10) Reference numerals in the accompanying drawings: electric wireless charging device **100**; charging dock **11**; fixed mount **111**; fixed part **1111**; guiding part **1112**; limit stop **112**; limit part **1121**; guiding member **113**; first limit guiding part **114**; second limit guiding part **115**; rotating charging base **12**; rotating support **121**; connecting part **1211**; guiding mating part **1212**; wire through hole **1213**; mobile phone charging part **122**; first driving component **13**; driving body **131**; driving shaft **132**; first charging part **14**; first toothed part **141**; first guiding sliding part **142**; accommodating cavity **143**; second charging part **15**; second toothed part **151**; second guiding sliding part **152**; second driving component **16**; driving main body **161**; driving gear **162**; watch magnetic member **17**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) In order to make those skilled in the art better understand the solutions of embodiments of the present disclosure, the technical solutions in the embodiments of the present disclosure are further described clearly and completely below in combination with the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are some embodiments of the present disclosure, but not all embodiments. All other embodiments obtained by the ordinary skilled in the art based on the embodiments of the present disclosure without creative efforts shall fall within the protection scope of the present disclosure.

(12) Technical solutions in the embodiments of the present disclosure are described clearly and completely below in combination with accompanying drawings in the embodiments of the present disclosure.

(13) Referring to FIG. 1, an embodiment of the present disclosure provides an electric wireless charging device **100**. The electric wireless charging device **100** may be used to charge electronic devices such as mobile phones, watches, and earphones, thereby meeting charging requirements of different electronic devices.

(14) Referring to FIG. 1 to FIG. 4 together, in some embodiments, the electric wireless charging device **100** includes a charging dock **11**, a rotating charging base **12**, and a first driving component **13**; the first driving component **13** includes a driving body **131** and a driving shaft **132**; the driving shaft **132** is connected with the charging dock **11**; the driving body **131** is connected with the rotating charging base **12**, and the driving body **131** is rotatably connected with the driving shaft **132**; and the rotation of the driving body **131** relative to the driving shaft **132** is suitable for driving the rotating charging base **12** to rotate.

(15) In this way, the electric wireless charging device **100** may charge different electronic devices respectively through the charging dock **11** and the rotating charging base **12**, which is helpful to enhance functionalities of the electric wireless charging device **100**. Furthermore, the driving body **131** may drive the rotating charging base **12** to rotate, whereby the rotating charging base **12** may automatically rotate to an appropriate position, which is convenient for a user to place the electronic device in an appropriate position for charging, and eliminates the need for the user to manually toggle the rotating charging base **12**, thereby improving user operation.

(16) Different electronic devices refer to different types of electronic devices. For example, the electric wireless charging device **100** may charge a mobile phone and a wireless rechargeable watch respectively through the charging dock **11** and the rotating charging base **12**. For another example, the electric wireless charging device **100** may charge a mobile phone and an earphone respectively through the charging dock **11** and the rotating charging base **12**. This may be set specifically according to an actual situation. Different electronic devices may also refer to the same type of electronic devices. For example, the electric wireless charging device **100** may charge different mobile phones respectively through the charging dock **11** and the rotating charging base **12**. For another example, the electric wireless charging device **100** may charge different earphones respectively through the charging dock **11** and the rotating charging base **12**. This may be set specifically according to an actual situation. The following description takes the electric wireless charging device **100** charging different types of electronic devices as an example.

(17) Exemplarily, the rotating charging base **12** may support the charging of a mobile phone. A magnet component and a mobile phone charging coil may be arranged in the rotating charging base **12**, and the user may place the mobile phone on the rotating charging base **12**, such as on the mobile phone charging part **122**, for charging; the charging dock **11** may support the charging of an earphone, an earphone charging coil may be arranged in the charging dock **11**, and the user may place the earphone on the charging dock **11**, such as on the second charging part **15**, for charging; and the charging dock **11** may further support the charging of a wireless rechargeable watch, a magnet may be arranged in the charging dock **11**, and the user may place the wireless rechargeable watch on the charging dock **11**, such as on the first charging part **14**, for charging.

(18) Referring to FIG. 3 and FIG. 4, in some embodiments, the charging dock **11** includes a fixed

mount **111**, the fixed mount **111** is provided with a fixed part **1111**, the driving shaft **132** is installed on the fixed part **1111**, the rotating charging base **12** includes a rotating support **121**, the rotating support **121** is provided with a connecting part **1211**, the driving body **131** is installed on the connecting part **1211**, and the rotation of the driving body **131** relative to the driving shaft **132** is suitable for driving the rotating support **121** to rotate.

(19) In this way, the fixed part **1111** may locate a position where the driving shaft **132** is connected with the fixed mount **111**, which is helpful to improve the accuracy of connection of the driving shaft **132** to the fixed mount **111**, further to improve the firmness of the connection between the driving shaft **132** and the fixed part **1111**, and further to reduce the situation that the driving shaft **132** is detached from the fixed part **1111**. The connecting part **1211** may locate a position where the driving body **131** is connected with the rotating support **121**, which is helpful to improve the accuracy of connection of the driving body **131** to the rotating support **121**, further to improve the firmness of the connection between the driving body **131** and the connecting part **1211**, and further to reduce the situation the driving body **131** is detached from the connecting part **1211**.

(20) Exemplarily, the fixed part **1111** may be a clamping hole, which allows the driving shaft **132** to be clamped in the clamping hole, thereby locating a connection position between the driving shaft **132** and the fixed mount **111**, and further improving the firmness of the connection between the driving shaft **132** and the fixed part **1111**; and the connecting part **1211** may be a slot, which allows the driving body **131** to be clamped in the slot, thereby locating a connection position between the driving body **131** and the rotating support **121**, and further improving the firmness of the connection between the driving body **131** and the connecting part **1211**.

(21) In some embodiments, the fixed mount **111** is also provided with a guiding part **1112**, the rotating support **121** is provided with a guiding mating part **1212**, and the guiding mating part **1212** is slidably connected with the guiding part **1112**.

(22) In this way, the guiding part **1112** may limit a rotating path of the rotating support **121**, the rotating support **121** may rotate along a set path through the sliding cooperation between the guiding mating part **1212** and the guiding part **1112**, thereby reducing the situation that the rotating support **121** deviates from the set path.

(23) The cooperation between the guiding part **1112** and the guiding mating part **1212** may be cooperation between a block and a groove. For example, the guiding part **1112** may be a guiding groove, the guiding groove may be located on a surface of the fixed mount **111** facing the rotating support **121**, the guiding mating part **1212** may be guiding block, the guiding block may be located on a surface of the rotating support **121** facing the fixed mount **111**, and the guiding block is slidably located in the guiding groove. In other embodiments, the cooperation between the guiding part **1112** and the guiding mating part **1212** may also be cooperation between blocks. This may be set specifically according to an actual situation.

(24) A central angle of the guiding part **1112** may be limited by setting a shape of the guiding part **1112**, thereby limiting a rotating angle of the rotating support **121**. For example, the guiding part **1112** may be a semicircular guiding groove, whereby the central angle of the guiding part **1112** is 180° , causing the rotating support **121** to rotate for 180° relative to the guiding part **1112**. For another example, the guiding part **1112** may be a quarter-circular guiding groove, whereby the central angle of the guiding part **1112** is 90° , causing the rotating support **121** to rotate for 90° relative to the guiding part **1112**. This may be set specifically according to an actual situation.

(25) In some embodiments, the guiding mating part **1212** is provided with a wire through hole **1213**, and the wire through hole **1213** penetrates through the guiding part **1112**.

(26) In this way, a wire of the electric wireless charging device **100** may be accommodated in the wire through hole **1213**, and electrically connected with a device in the charging dock **11** and a device in the rotating charging base **12** respectively to ensure the normal work of each device, thereby facilitating the reasonable layout of the wires, and reducing the situation that the wires get tangled due to the scattering of the wires in the charging dock **11** and the rotating charging base **12**.

Because the wire may directly penetrate through the wire through hole **1213** to be electrically connected with the device in the charging dock **11** and the device in the rotating charging base **12**, the fixed mount **111** and the rotating support **121** may not hinder the assembly of the wires; and the wires can be electrically connected with the devices at two sides of the rotating support **121** without bypassing the peripheries of the fixed mount **111** and rotating support **121**, thereby improving the structural compactness of the electric wireless charging device **100**, and further facilitating the layout of the wires.

(27) Referring to FIG. 3 to FIG. 5 together, in some embodiments, the charging dock **11** further includes a limit stop **112** and guiding members **113**, the limit stop **112** is connected with the fixed mount **111**, the limit stop **112** is provided with a limit part **1121**, the guiding members **113** penetrate through the limit part **1121** and are connected with the rotating support **121**, an end, facing away from the rotating support **121**, of each guiding member **113** is limited in the limit part **1121**, the guiding members **113** are suitable for limiting the movement of the rotating support **121** along an axial direction of the guiding members **113**, and the guiding members **113** and the rotating support **121** rotate synchronously around a rotating axis of the driving body **131**.

(28) In this way, the guiding members **113** may limit the movement of the rotating support **121** along the axial direction of the guiding members **113**, the guiding members **113** may also support the rotation of the rotating support **121** around the rotating axis of the driving body **131**, to realize the rotatable connection between the rotating support **121** and the limit stop **112**, thereby reducing the situation that the rotating support **121** moves along the axial direction of the guiding members **113** when rotating, and improving the rotating stability of the rotating support **121**.

(29) The guiding member **113** may be a screw and other structures, the limit part **1121** may be a through hole, a screw stem of the screw penetrates through the limit part **1121** and is connected with the rotating support **121**, a head of the screw is limited in the limit part **1121**, and the screw stem and head of the screw are located at two sides of the limit part **1121** respectively, whereby the screw may limit the movement of the rotating support **121** along the axial direction of the screw, and the head of the screw may also rotate around the end of the limit part **1121** along the rotating axis of the driving body **131**.

(30) In some embodiments, there are a plurality of guiding members **113**, the plurality of guiding members **113** are disposed around the periphery of the driving body **131**, and the plurality of guiding members **113** are distributed symmetrically relative to the rotating axis of the driving body **131**.

(31) In this way, it is helpful to improve the firmness of rotatable connection between the limit stop **112** and the rotating support **121**, better to reduce the situation that the rotating support **121** moves along the axial direction of the guiding member **113**, and also better to improve the rotating stability of the rotating support **121** around the rotating axis of the driving body **131**.

(32) For example, the quantity of the guiding members **113** may be two, three, four, five, six, seven, eight, or even more, and this may be set specifically according to an actual situation.

(33) Referring to FIG. 1 and FIG. 2 again, in some embodiments, the electric wireless charging device **100** includes a first charging part **14** and a second charging part **15**, and the first charging part **14** and the second charging part **15** are both installed on the charging dock **11**.

(34) In this way, the charging dock **11** may charge two electronic devices through the first charging part **14** and the second charging part **15**, which is helpful to enhance functionalities of the charging dock **11**, and further to enhance functionalities of the electric wireless charging device **100**.

(35) The first charging part **14** and the second charging part **15** may be of a same structure so as to charge the same type of electronic devices; and the first charging part **14** and the second charging part **15** may further be of different structures, so as to charge different types of electronic devices. This may be set specifically according to an actual situation. The following description takes the first charging part **14** and the second charging part **15** that are of different structures for charging different types of electronic devices as an example.

(36) In some embodiments, the first charging part **14** and the second charging part **15** are located on two opposite ends of the charging dock **11** respectively.

(37) In this way, it is helpful for the user to charge two electronic devices from two ends of the charging dock **11**, thereby reducing the mutual interference of the first charging part **14** and the second charging part **15** when in work, and ensuring the normal work of the first charging part **14** and the second charging part **15**.

(38) Referring to FIG. 1, FIG. 2, FIG. 6, and FIG. 7 again, in some embodiments, the electric wireless charging device **100** includes a second driving component **16**, the second driving component **16** includes a driving main body **161** and a driving gear **162**, the first charging part **14** is provided with a first toothed part **141**, the second charging part **15** is provided with a second toothed part **151**, the first toothed part **141** is engaged with one end of the driving gear **162**, the second toothed part **151** is engaged with the other end of the driving gear **162**, the driving main body **161** is suitable for driving the driving gear **162** to rotate, the rotation of the driving gear **162** is suitable for driving the first toothed part **141** to move towards a first direction, the rotation of the driving gear **162** is further suitable for driving the second toothed part **151** to move towards a second direction, where the first direction and the second direction are different.

(39) In this way, the driving main body **161** may drive the driving gear **162** to rotate, and the rotation of the driving gear **162** may drive the first toothed part **141** and the second toothed part **151** to move in different directions, whereby the first charging part **14** and the second charging part **15** may move automatically to an appropriate position, which is convenient for the user to place the electronic device in the appropriate position for charging; and the user may charge the electronic device without manually pushing and pulling the first charging part **14** and the second charging part **15**, thereby facilitating the operation of the user.

(40) Furthermore, the charging dock **11** is provided with the second driving component **16** to realize the automatic movement of the first charging part **14** and the second charging part **15**, thereby simplifying the structure of the charging dock **11**, and saving the manufacturing cost.

(41) "Different" refers to that the first direction is different from the second direction, for example, the first direction is opposite to the second direction.

(42) Exemplarily, when the driving main body **161** drives the driving gear **162** to rotate clockwise, the clockwise rotation of the driving gear **162** may drive the first toothed part **141** and the second toothed part **151** to extend out of the charging dock **11**, as shown in FIG. 2, whereby the first charging part **14** and the second charging part **15** may move automatically to an appropriate position; and the first charging part **14** and the second charging part **15** may extend out of the charging dock **11** without the manual pulling by the user, thereby facilitating the user to place the electronic devices on the first charging part **14** and the second charging part **15** for charging. When the driving main body **161** drives the driving gear **162** to rotate counterclockwise, the counterclockwise rotation of the driving gear **162** may drive the first toothed part **141** and the second toothed part **151** respectively to extend into the charging dock **11**, whereby the first charging part **14** and the second charging part **15** may move automatically into the charging dock **11**, as shown in FIG. 1, thereby facilitating the restoration and accommodation of the first charging part **14** and the second charging part **15**; and the first charging part **14** and the second charging part **15** may extend into the charging dock **11** without the manual pushing by the user, thereby facilitating the operation of the user.

(43) Referring to FIG. 6 and FIG. 7 together, in some embodiments, the charging dock **11** is provided with a first limit guiding part **114** and a second limit guiding part **115**, the first charging part **14** is provided with a first guiding sliding part **142**, the first guiding sliding part **142** is slidably connected with the first limit guiding part **114**, the second charging part **15** is provided with a second guiding sliding part **152**, and the second guiding sliding part **152** is slidably connected with the second limit guiding part **115**.

(44) In this way, the first limit guiding part **114** and the second limit guiding part **115** may limit

movement paths of the first guiding sliding part **142** and the second guiding sliding part **152** respectively, whereby the first charging part **14** and the second charging part **15** may move along a set path, which is helpful to reduce the situation that the first charging part **14** and the second charging part **15** deviate from the set path, further to reduce the situation that the first charging part **14** and the second charging part **15** cannot extend out of the charging dock **11** due to excessively small displacement of the first charging part **14** and the second charging part **15**, and further to reduce the situation that the first charging part **14** and the second charging part **15** are detached from the charging dock **11** due to excessively large displacement.

(45) The cooperation between the first guiding sliding part **142** and the first limit guiding part **114** may be cooperation between a block and a groove, for example, the first guiding sliding part **142** may be a guiding groove, the guiding groove is located on a surface of the first charging part **14** facing the charging dock **11**, the first limit guiding part **114** may be a guiding block, the guiding block is located on a surface of the charging dock **11** facing the first charging part **14**, and the guiding block is slidably located in the guiding groove. In other embodiments, the cooperation between the first guiding sliding part **142** and the first limit guiding part **114** may also be cooperation between blocks. This may be set specifically according to an actual situation.

(46) The cooperation between the second guiding sliding part **152** and the second limit guiding part **115** may be cooperation between a block and a groove, for example, the second guiding sliding part **152** may be a guiding groove, the guiding groove is located on a surface of the second charging part **15** facing the charging dock **11**, the second limit guiding part **115** may be a guiding block, the guiding block is located on a surface of the charging dock **11** facing the second charging part **15**, and the guiding block is slidably located in the guiding groove. In other embodiments, the cooperation between the second guiding sliding part **152** and the second limit guiding part **115** may also be cooperation between blocks. This may be set specifically according to an actual situation.

(47) Referring to FIG. **6** and FIG. **8** together, in some embodiments, the electric wireless charging device **100** further includes a watch magnetic member **17**, the first charging part **14** is provided with an accommodating cavity **143**, and the watch magnetic member **17** is installed in the accommodating cavity **143** in a reversible manner.

(48) Due to variations in the orientation of north and south poles of a magnet inside different types of watches, the watch magnetic member **17** is installed in the accommodating cavity **143** in a reversible manner, whereby the watch magnetic member **17** may be adapted to the corresponding north and south poles of the magnet of different watches, thereby matching the orientations of magnetic poles of magnets in different watches for charging. This design eliminates the need to replace a charging module of the first charging part **14** to adapt to the charging for different watches, which is helpful to improve the adaptability of the first charging part **14**, and the charging flexibility of the first charging part **14**.

(49) In conclusion, according to the electric wireless charging device **100** provided by the embodiments of the present disclosure, the first driving component **13** of the electric wireless charging device **100** includes the driving body **131** and the driving shaft **132**, the driving shaft **132** is connected with the charging dock **11**, the driving body **131** is connected with the rotating charging base **12**, the driving body **131** is rotatably connected with the driving shaft **132**, and the rotation of the driving body **131** relative to the driving shaft **132** is suitable for driving the rotating charging base **12** to rotate. In this way, the electric wireless charging device **100** may charge different electronic devices respectively through the charging dock **11** and the rotating charging base **12**, which is helpful to enhance functionalities of the electric wireless charging device **100**. Furthermore, the driving body **131** may drive the rotating charging base **12** to rotate, whereby the rotating charging base **12** may automatically rotate to an appropriate position, which is convenient for the user to place the electronic device in an appropriate position for charging, and eliminates the need for the user to manually toggle the rotating charging base **12**, thereby facilitating user operation.

(50) Additionally, terms “first” and “second” are only for the purpose of description, and cannot be understood as specific indication or special structure. The description of the term “some embodiments” means specific features, structures, materials, or characteristics described in conjunction with the embodiment or example are contained in at least one embodiment or example of the present disclosure. In the present disclosure, the schematic representations of the above terms are not necessary for the same embodiment or example. Moreover, the described specific features, structures, materials or characteristics may be combined in an appropriate way in any or more embodiments or examples. Furthermore, different embodiments or examples and features of different embodiments or examples described in the present disclosure may be combined by those skilled in the art without contradicting each other.

(51) The above embodiments are only used for illustrating the technical solutions of the present disclosure rather than limiting the present disclosure. Although the present disclosure is described in detail with reference to the foregoing embodiments, it should be understood by those skilled in the art that the technical solutions described in the foregoing embodiments may be modified or some technical features may be equivalently substituted. However, these modifications or substitutions do not make the essence of the corresponding technical solutions depart from the spirit and scope of the technical solutions of various embodiments of the present disclosure, and should fall within the protection scope of the present disclosure.

Claims

1. An electric wireless charging device, comprising: a charging dock and a rotating charging base; and a first driving component, the first driving component comprising a driving body and a driving shaft, the driving shaft being connected with the charging dock, the driving body being connected with the rotating charging base, the driving body being rotatably connected with the driving shaft, and the rotation of the driving body relative to the driving shaft being suitable for driving the rotating charging base to rotate; a first charging part and a second charging part, the first charging part and the second charging part being both installed on the charging dock, and the first charging part and the second charging part being located at two opposite ends of the charging dock respectively; and a second driving component, the second driving component comprising a driving main body and a driving gear, the first charging part being provided with a first toothed part, the second charging part being provided with a second toothed part, the first toothed part being engaged with one end of the driving gear, the second toothed part being engaged with the other end of the driving gear, the driving main body being suitable for driving the driving gear to rotate, the rotation of the driving gear being suitable for driving the first toothed part to move towards a first direction, the rotation of the driving gear being further suitable for driving the second toothed part to move towards a second direction, wherein the first direction and the second direction are different, the charging dock comprises a fixed mount, the fixed mount is provided with a fixed part, the driving shaft is installed on the fixed part, the rotating charging base comprises a rotating support, the rotating support is provided with a connecting part, the driving body is installed on the connecting part, and the rotation of the driving body relative to the driving shaft is suitable for driving the rotating support to rotate, wherein the charging dock further comprises a limit stop and guiding members, the limit stop is connected with the fixed mount, the limit stop is provided with a limit part, the guiding members penetrate through the limit part and are connected with the rotating support, an end, facing away from the rotating support, of each guiding member is limited in the limit part, the guiding members are suitable for limiting the movement of the rotating support along an axial direction of the guiding members, and the guiding members and the rotating support rotate synchronously around a rotating axis of the driving body.

2. The electric wireless charging device according to claim 1, wherein the fixed mount is further provided with a guiding part, the rotating support is provided with a guiding mating part, and the

guiding mating part is slidably connected with the guiding part.

3. The electric wireless charging device according to claim 2, wherein the guiding mating part is provided with a wire through hole, and the wire through hole penetrates through the guiding part.

4. The electric wireless charging device according to claim 1, wherein there are a plurality of guiding members, the plurality of guiding members are disposed around the periphery of the driving body, and the plurality of guiding members are distributed symmetrically relative to the rotating axis of the driving body.

5. The electric wireless charging device according to claim 1, wherein the charging dock is provided with a first limit guiding part and a second limit guiding part, the first charging part is provided with a first guiding sliding part, the first guiding sliding part is slidably connected with the first limit guiding part, the second charging part is provided with a second guiding sliding part, and the second guiding sliding part is slidably connected with the second limit guiding part.

6. The electric wireless charging device according to claim 1, wherein the electric wireless charging device comprises the first charging part and a watch magnetic member, the first charging part is provided with an accommodating cavity, and the watch magnetic member is installed in the accommodating cavity in a reversible manner.
